



DIGITAL FORENSICS IN MOBILE PHONES-SEC 4336-3

Chapter 1: Mobile Device Fundamentals

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Chapter Contents



Hardware Architecture

SoC, CPU/GPU, baseband, sensors



Memory & Storage

RAM vs. NAND flash, eMMC/UFS



Mobile Operating Systems

Android vs. iOS design philosophy



OS Architecture Layers

Kernel to application stack



File Systems

EXT4, F2FS, APFS — why they matter



Identifiers & SIM/Network

IMEI, IMSI, ICCID, UDID



Boot & Security Model

Secure boot, encryption, lock screen



Forensic Implications

Connecting fundamentals to evidence

Why Start With Fundamentals?

Every forensic decision — which acquisition method to use, whether encryption can be bypassed, where deleted data might still live — depends on understanding what is actually inside the device.



Acquisition Strategy

Chip-off, JTAG, or logical extraction depend on the SoC, storage type, and boot security in place.



Where Evidence Lives

OS architecture and file system determine where app data, logs, and deleted records are physically stored.



What You Can Bypass

Encryption and secure boot models define the legal and technical limits of what an examiner can access.

Mobile Device Hardware Architecture

A modern smartphone packs a full computer into a sealed, battery-powered shell. Six subsystems matter most to a forensic examiner:



System-on-Chip (SoC)

CPU, GPU, baseband modem, and secure enclave fused onto one die.



RAM (Volatile)

Working memory — cleared on power-off; live acquisition target.



NAND Flash Storage

Non-volatile storage (eMMC/UFS) holding the OS, apps, and user data.



SIM / eSIM

Carrier identity module storing IMSI, contacts, and network credentials.



Baseband & Radios

Cellular, Wi-Fi, Bluetooth, GPS, NFC — separate processor from the main OS.



Sensors & Biometrics

Fingerprint/face sensors, accelerometer, gyroscope — often behind secure hardware.

Inside the System-on-Chip

CPU (Application Processor)

Runs the OS and apps — typically ARM big.LITTLE cores balancing power and performance.

GPU

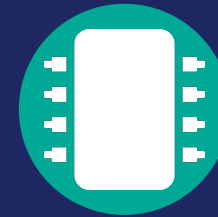
Handles graphics rendering and increasingly on-device machine learning workloads.

Baseband Processor

Manages cellular communication on its own firmware — a distinct attack surface from the main OS.

Secure Enclave / TEE

Isolated environment for keys, biometrics, and encryption — the primary obstacle to forensic bypass.



SoC

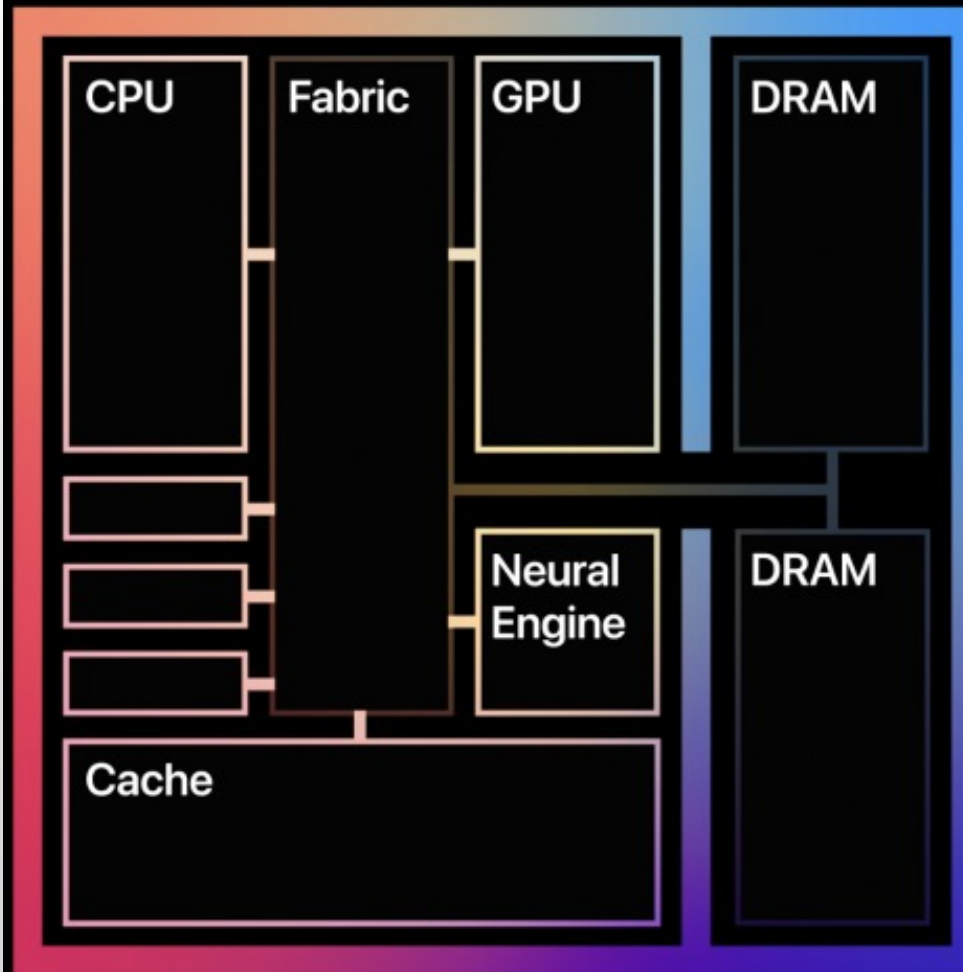
CPU Cores

GPU

Baseband Modem

Secure Enclave / TEE

- CPU needs an ecosystem of other chips to become a computer. They need memory chips for data, audio chips, graphics processors, connectivity chips, and more. **SoC, on the other hand, as the name suggests, has everything on a single chip and is more efficient.**
- **Apple has done precisely that with its M1 — 16 billion transistors that do everything a modern computer needs to do.** In comparison, a CPU still needs more chips around it to make a computer work — and that creates constraints for the machines.



Unified memory architecture

High bandwidth, low latency
Apple-designed package
Accessible to entire SoC

Storage & Memory: What Survives Power-Off



RAM — Volatile Memory

- Cleared instantly when power is lost
- Holds running processes, open app data, passwords, and decryption keys in-use
- Requires live acquisition — device must stay powered on
- Captured via specialized tools (e.g., JTAG live memory dump) before shutdown



NAND Flash — Non-Volatile

- Persists after power-off — the OS, apps, photos, and databases live here
- Implemented as eMMC (older/budget) or UFS (modern, faster) chips
- Primary target for physical/logical acquisition
- Wear-leveling and TRIM can complicate recovery of deleted data

Android vs. iOS: Two Design Philosophies



Android (Google)

Linux-kernel based, open-source, highly customizable by OEMs

Kernel	Modified Linux kernel
File System	EXT4 or F2FS
App Sandbox	Per-app Linux UID isolation
App Format	APK (ZIP-based package)
Fragmentation	Many OEM skins & versions

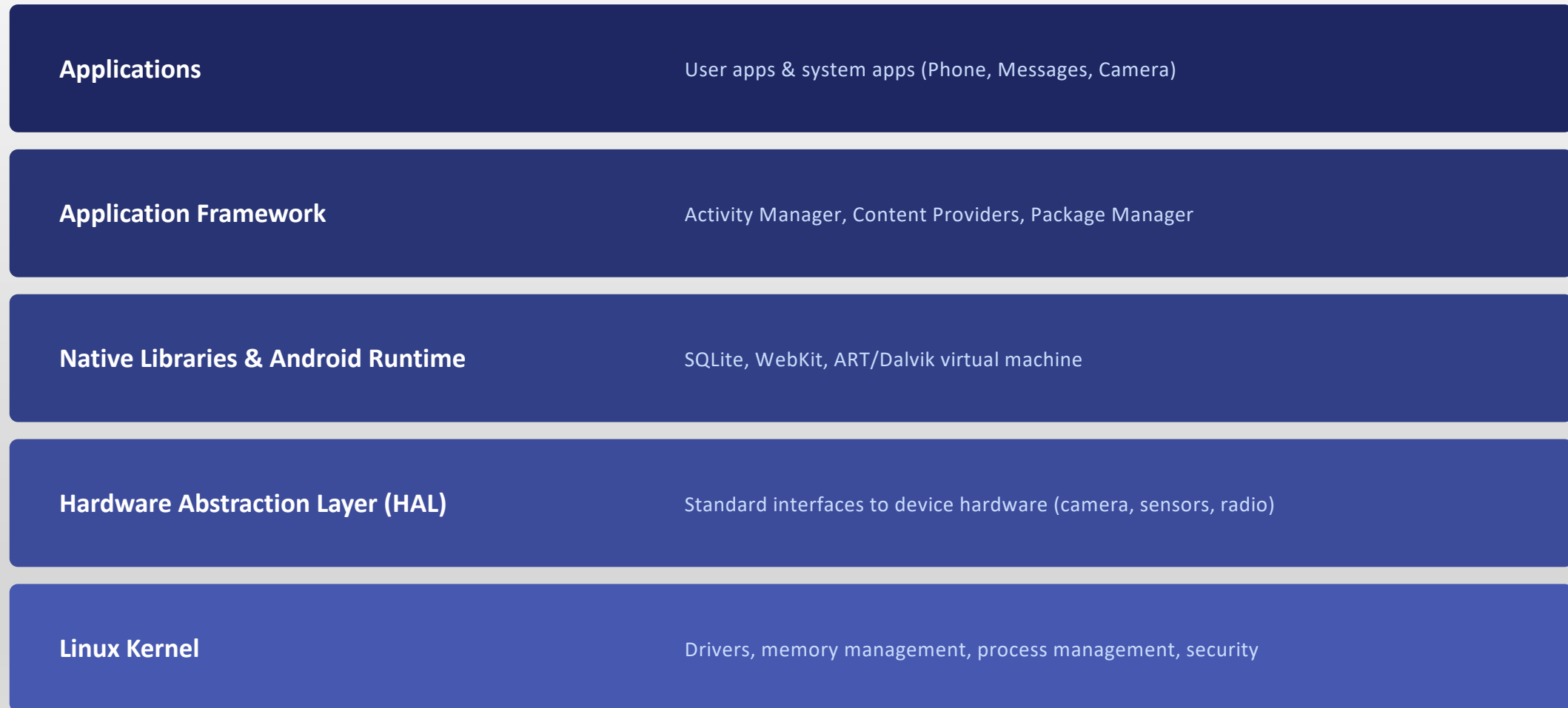


iOS (Apple)

Darwin/XNU-based, closed ecosystem, tight hardware-software integration

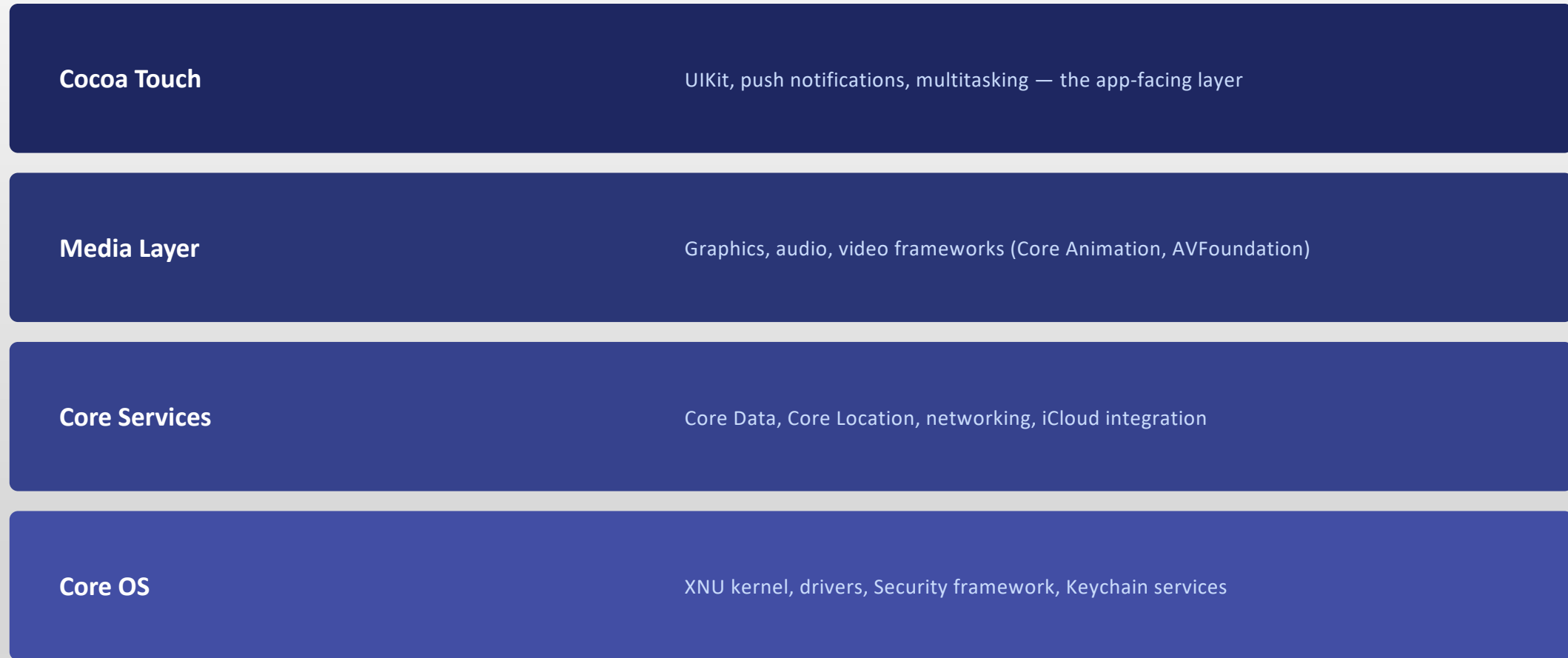
Kernel	XNU (hybrid Mach + BSD)
File System	APFS (formerly HFSX)
App Sandbox	Strict per-app container
App Format	IPA package
Fragmentation	Single vendor, unified updates

Android: A Layered Stack



Forensic data of interest concentrates at the top three layers — application and framework data.

iOS: A Layered Stack



The Keychain (Core OS layer) stores credentials and tokens — a high-value forensic artifact.

File Systems and Why They Matter

The file system governs how deleted data is marked, overwritten, and — sometimes — recoverable.

EXT4

Android (legacy/mid-range)

Journaling file system; deleted file metadata often briefly recoverable before journal wraps.

F2FS

Android (modern flash-optimized)

Designed for NAND flash; log-structured writes complicate traditional carving techniques.

APFS

iOS, iPadOS, modern macOS

Copy-on-write, native encryption per-file; snapshots can preserve prior states of data.

Identifiers That Tie a Device to a Case



IMEI

International Mobile Equipment Identity — unique to the physical handset; tracks device across SIM changes.



IMSI

International Mobile Subscriber Identity — unique to the SIM, identifies the subscriber on the carrier network.



ICCID

Integrated Circuit Card ID — the SIM card's own serial number, printed on the card and stored on it.



UDID / Serial Number

Apple's Unique Device Identifier or Android hardware serial — ties acquisition data to one specific unit.

SIM Cards & Cellular Network Basics



What a SIM/USIM Stores

- IMSI (subscriber identity)
- ICCID (card serial number)
- Authentication keys (Ki)
- Contacts (limited entries)
- SMS messages (on older/legacy SIMs)
- Last dialed numbers (some cards)



Network Generations

GSM (2G)

Circuit-switched voice/SMS; basic call/SMS metadata

UMTS (3G)

Adds packet-switched data; early mobile internet

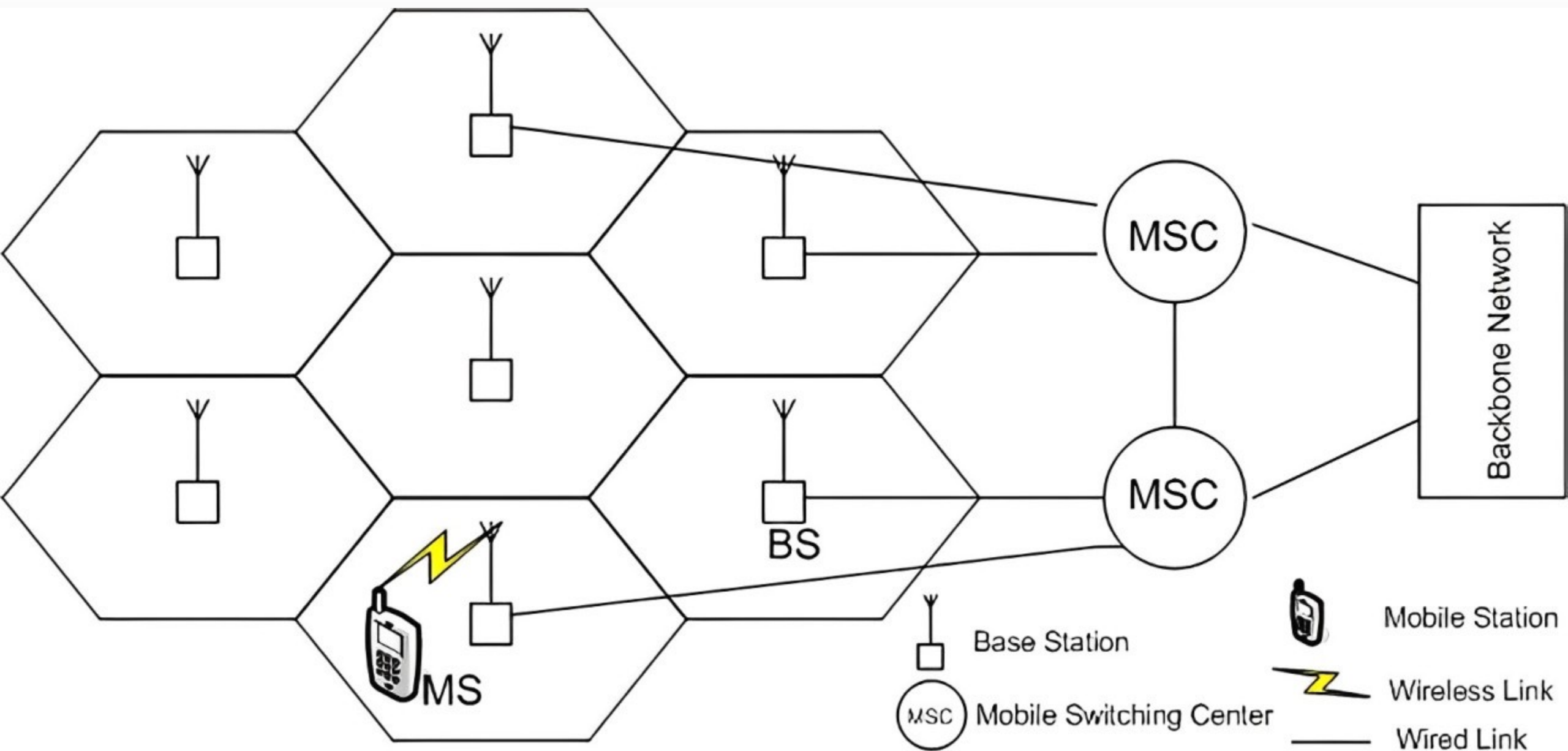
LTE (4G)

All-IP network; VoLTE calling, richer app data

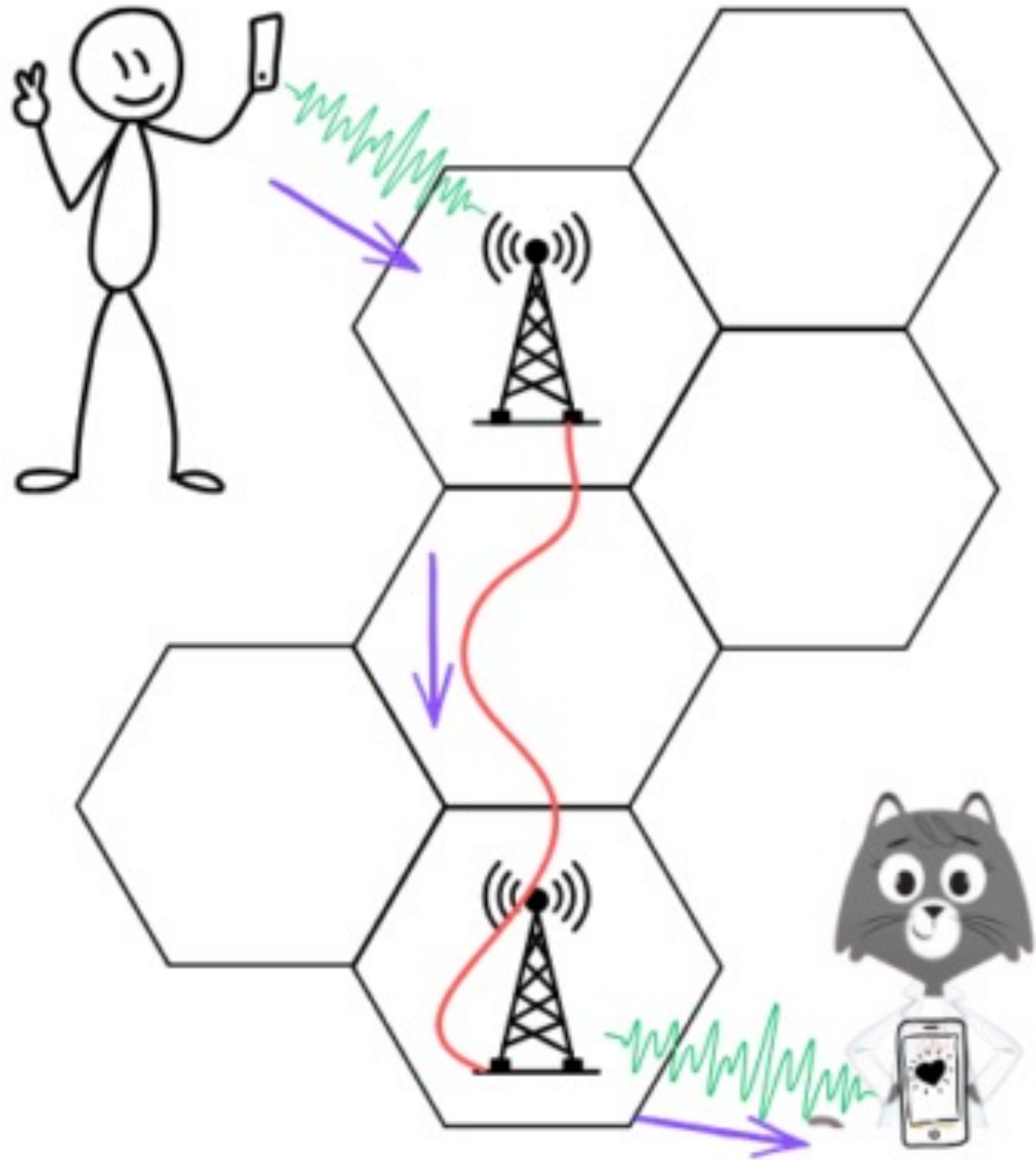
5G NR

High-speed data; network-based location precision

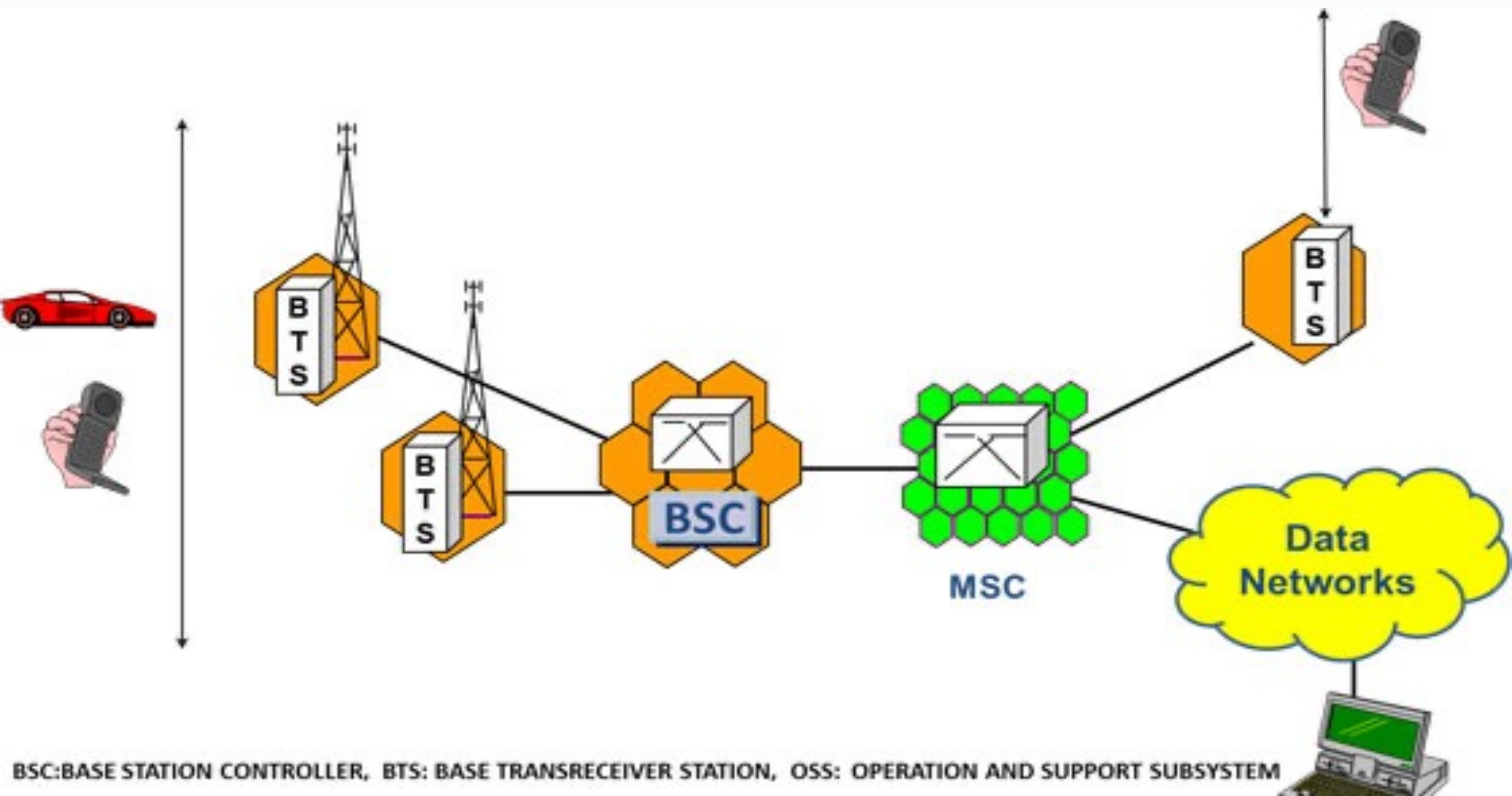
2G- NETWORK ARCHITECTURE



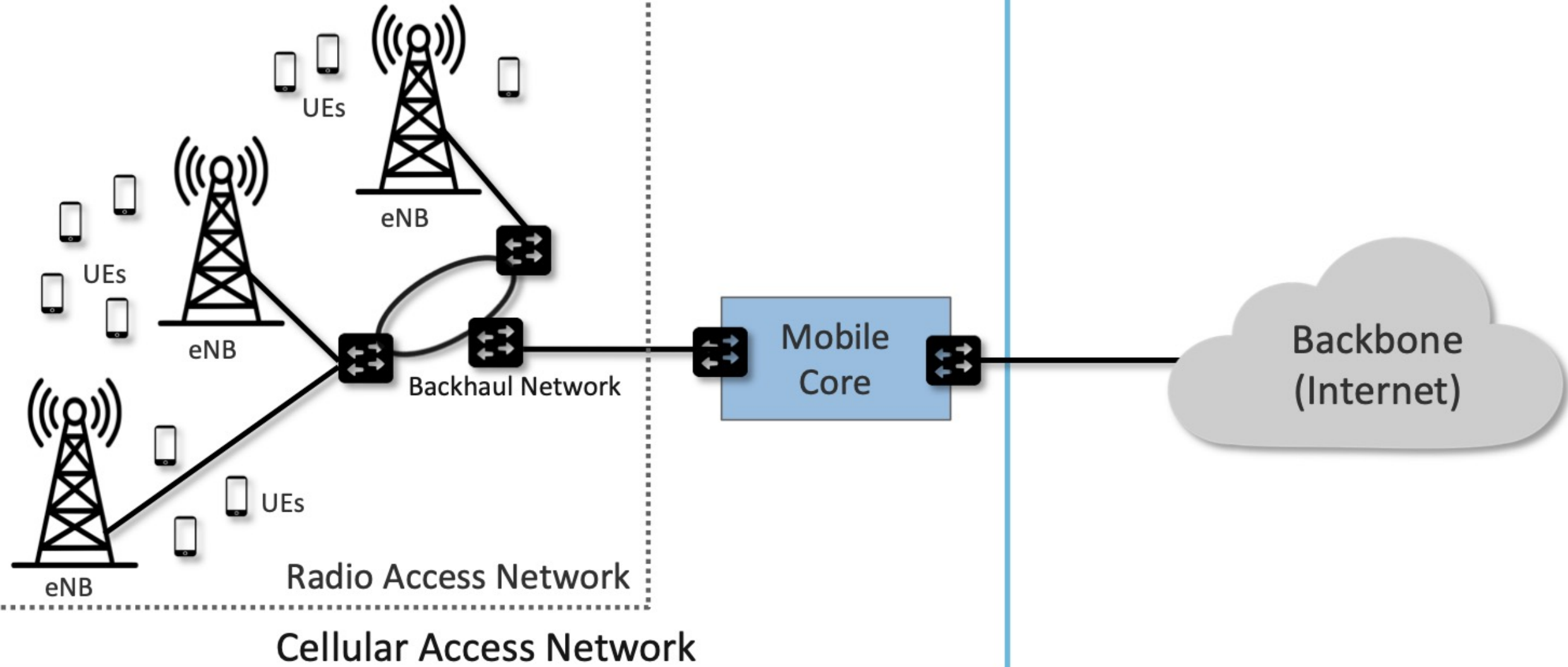
2G- NETWORK ARCHITECTURE



2G- NETWORK ARCHITECTURE



5G- NETWORK ARCHITECTURE



Connecting Fundamentals to evidence

Communication and network example

F Fundamentals

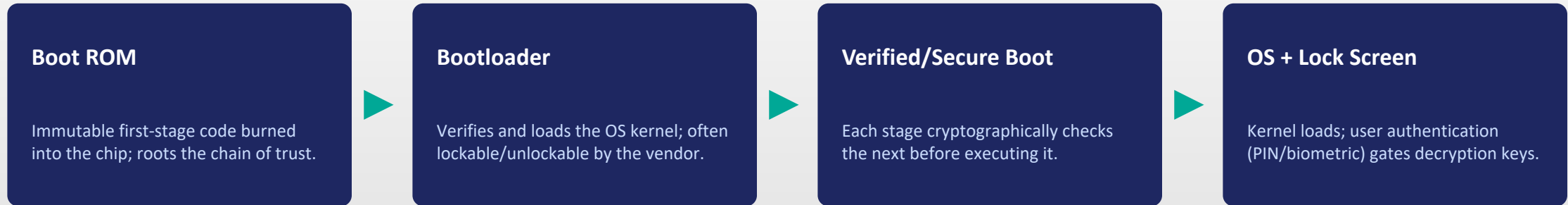
- Cellular network registration ties a device to towers over time
- Wi-Fi and Bluetooth connection logs record nearby networks/devices
- App-based messaging often relies on cloud sync, not just local storage
- Metadata accompanies nearly every communication event



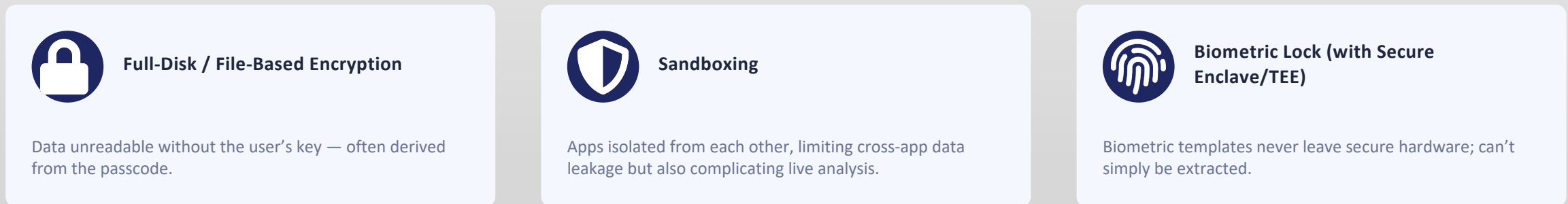
E Forensic Implications

- Cell tower and network logs support geolocation and timeline evidence
- Connection history corroborates or contradicts a suspect's account of movement
- Cloud sync means evidence may exist off-device, requiring separate legal process
- Metadata (timestamps, senders, routing) is often more court-durable than content itself

Boot Process & Security Features



Security Features That Shape an Investigation



Key Takeaways

MOBILE PHONE ARCHITECTURE

A mobile phone is a smart electronic device that accepts input, processes data, stores it, and provides output while enabling communication.

1. MAIN COMPONENTS

1. PROCESSOR (SoC)
The brain of the phone. Handles all processing and control.

2. MEMORY
Stores data and running apps. (RAM + ROM)

3. MODEM
Enables mobile network connection (2G / 3G / 4G / 5G)

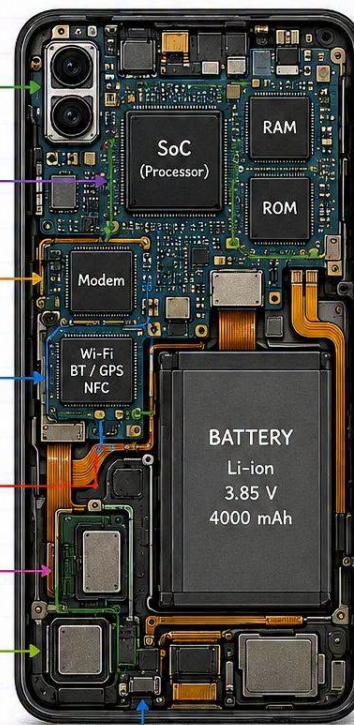
4. CONNECTIVITY
Wi-Fi, Bluetooth, GPS, NFC, and other wireless technologies.

5. DISPLAY
Shows visual output. (Touchscreen + Display Panel)

6. CAMERA
Captures photos and videos.

7. AUDIO SYSTEM
Includes speaker, microphone, and audio codec.

8. BATTERY
Provides power to all components.



2. HOW IT WORKS

INPUT
Touch, buttons, microphone, camera, sensors collect data.

PROCESSING
The processor (SoC) processes the data using instructions (apps & system).

OUTPUT
The display, speaker, vibration, and LEDs provide results to the user.

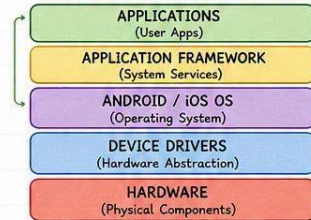
STORAGE
Data, apps, photos, videos, and files are saved in memory (storage).

COMMUNICATION
The modem and connectivity chips send and receive data (calls, internet, messages).

3. SENSORS

- Proximity Sensor – Detects nearby objects
- Ambient Light Sensor – Adjusts screen brightness
- Accelerometer – Detects movement and orientation
- Gyroscope – Detects rotation and motion
- Fingerprint Sensor – Security unlock
- Magnetometer – Detects magnetic field (compass)

4. SOFTWARE LAYERS



5. EXAMPLES OF USE

- Making calls
- Browsing the internet
- Sending messages
- Taking photos & videos
- Listening to music
- Playing games
- Navigation (GPS)
- Mobile payments

IN SHORT

A mobile phone architecture combines hardware and software components working together to take input, process it, give output, store data, and communicate — making our daily tasks easier and faster.



Key Takeaways

- Hardware architecture (SoC, RAM vs. NAND) determines whether live or dead acquisition is possible.
- OS design (Android's openness vs. iOS's sandboxing) shapes where and how you can legally and technically extract data.
- File systems dictate the odds of recovering deleted evidence — and which tools/techniques apply.
- Identifiers (IMEI, IMSI, ICCID, UDID) are essential for chain-of-custody and linking evidence to a specific device.
- Boot security and encryption define the technical and legal boundary of what an examiner can access without specialized authority.